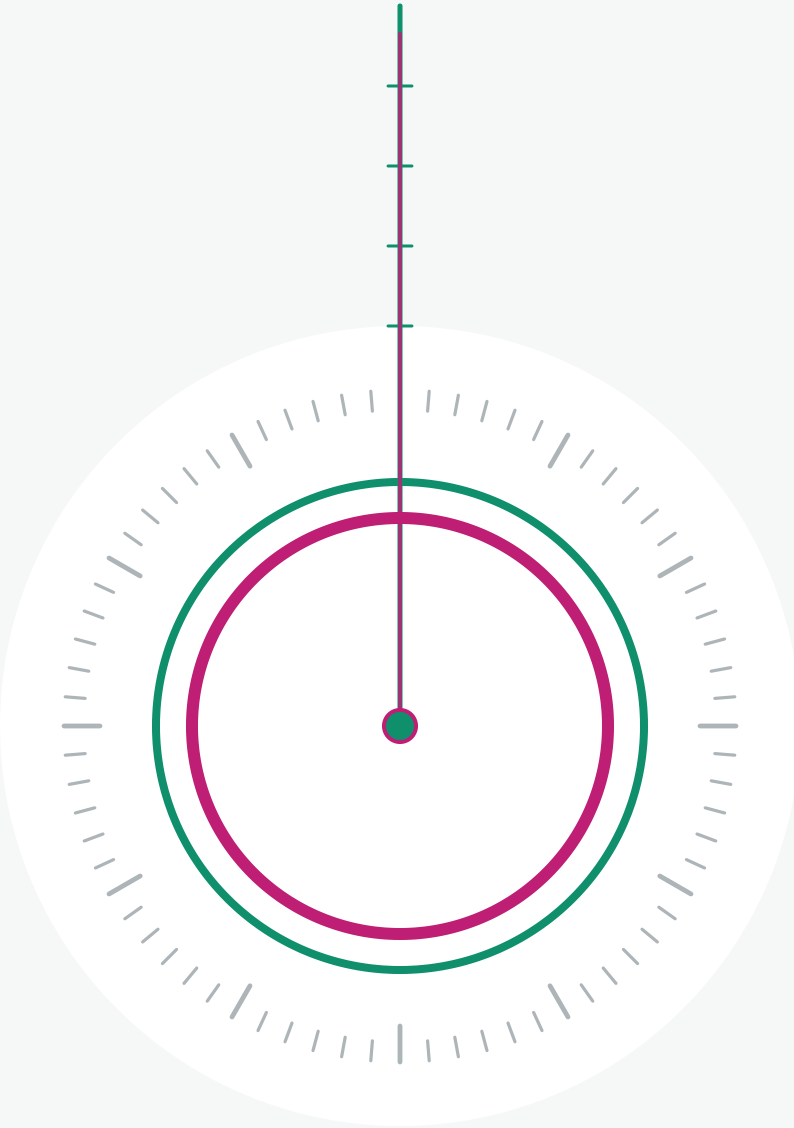
DATUM LINE

Every lab's cells. One "normal".

Membership to a shared repository
of every lab's control cells.



Kat Titterton
Cofounder, CSO



resume ↗

Kat Titterton

2026 — Axiom — founding technical staff

2023 — insitro — associate scientist III

2022 — MIT — technical associate II

2021 — abbvie — associate scientist II

2018 — MGH — research technician I

2017 — COLGATE — B.A. cellular neuroscience

2013

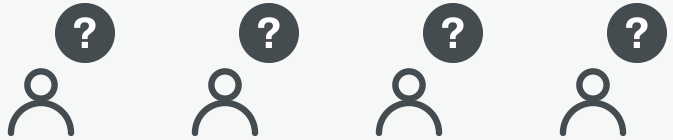
8
peer-reviewed papers
two in Nature Communications ↗

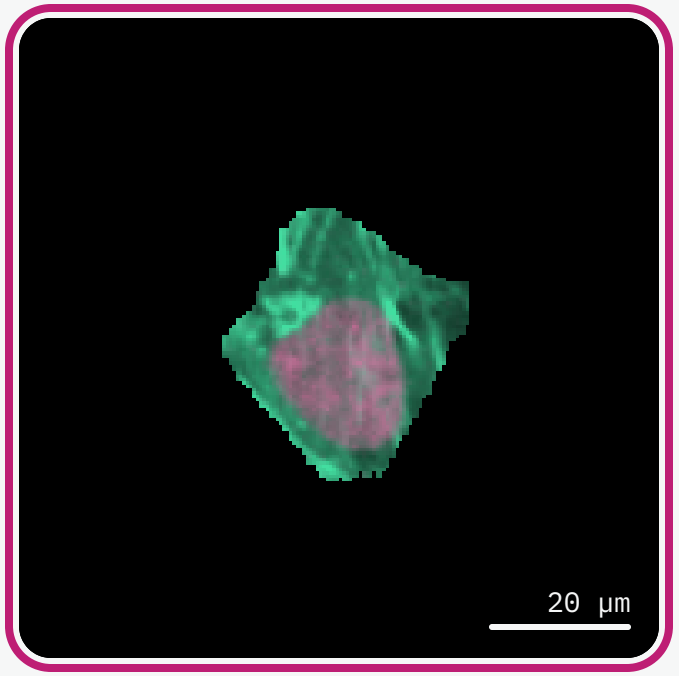
2
patent applications
filed at Axiom Bio

100,000+
compounds imaged
two Cell Painting datasets ↗

Do my cells look “normal”?

“normal” = published images + past data





my cells, lab 2

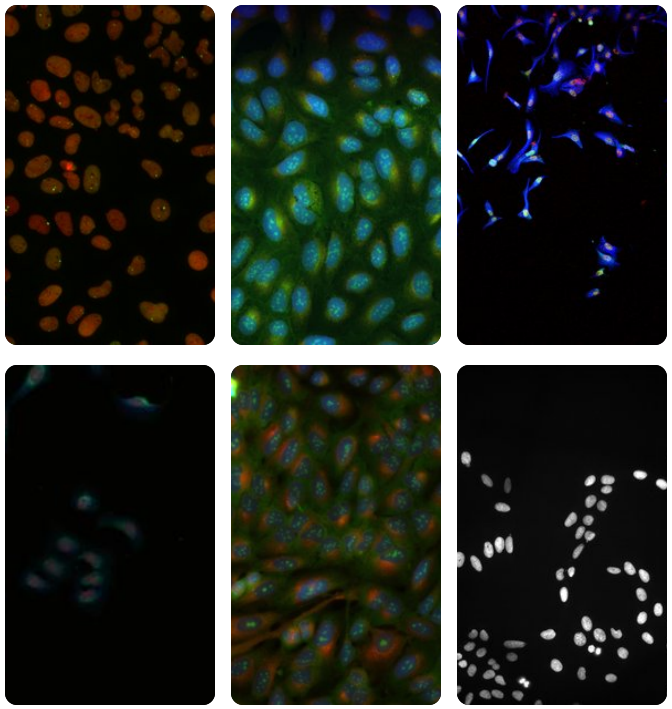
DMSO, no drug



Today: search, ask, guess.

search

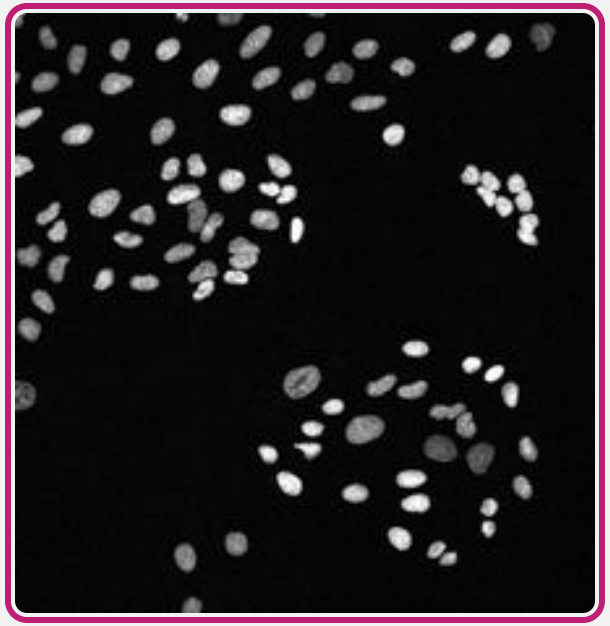
U2OS cells “normal”



ask

To: the paper’s authors

do my cells look right?



guess

RUN IT

$100,000s

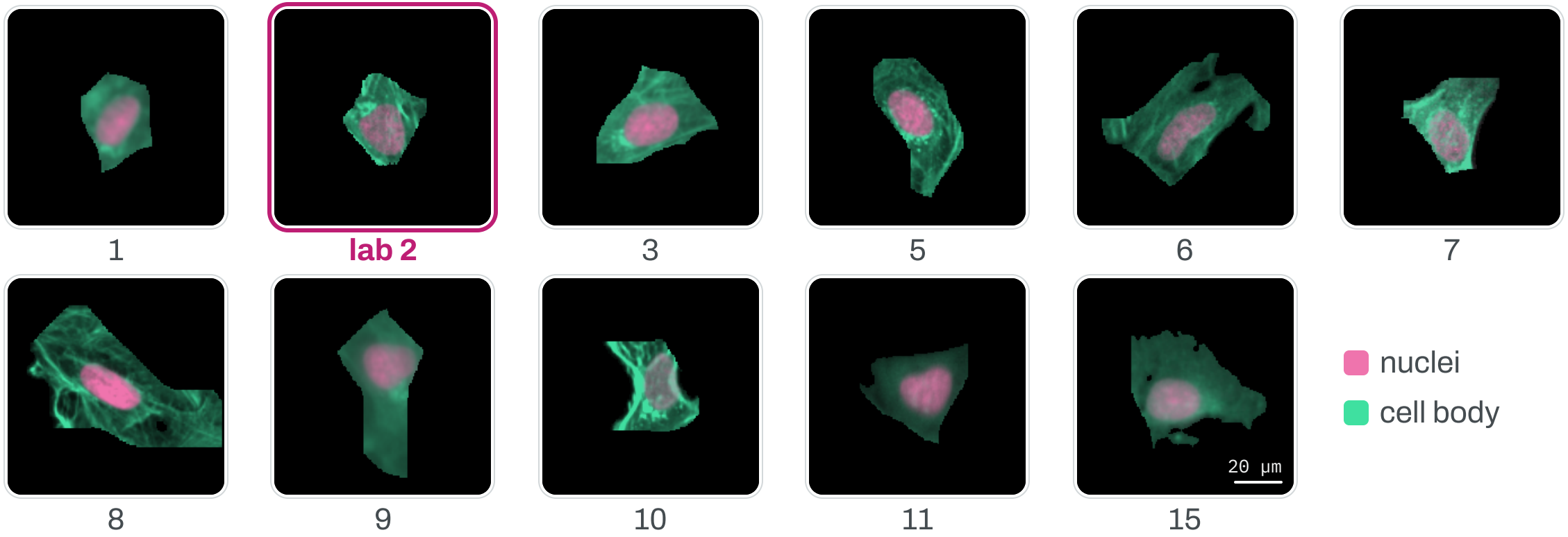
at risk

WAIT

months

lost

Each lab judges “normal” alone.



WHY NOW

2023 1.8B cells go public

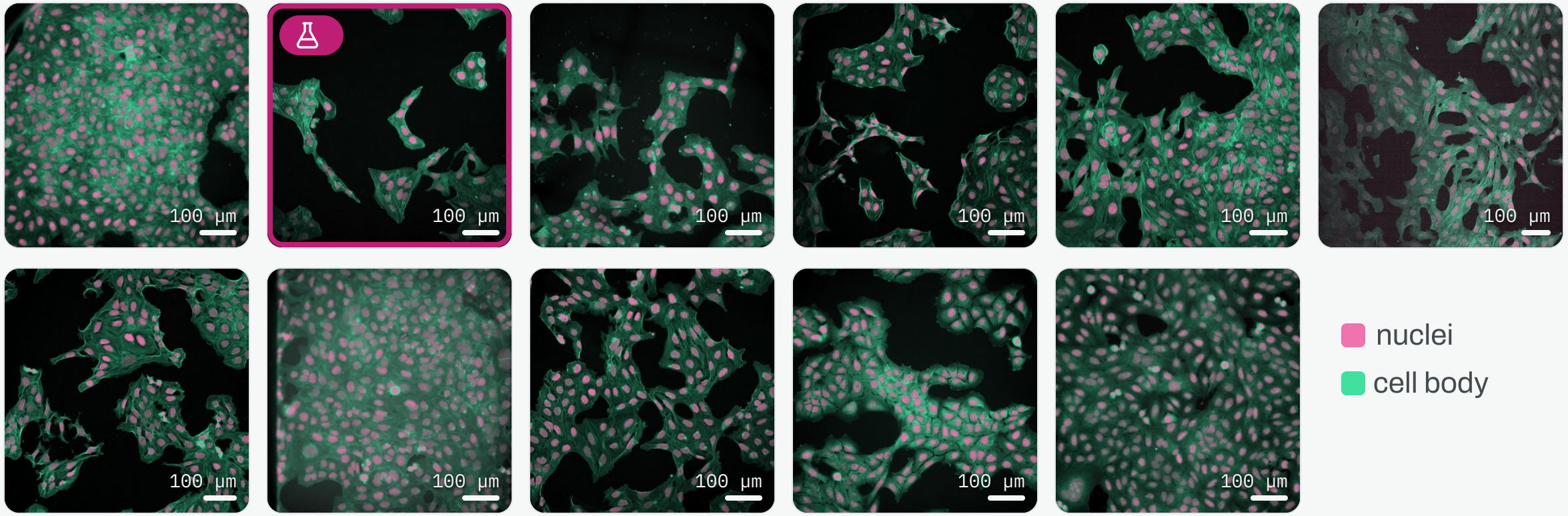
2026 $1.8B push: Google, Meta join

now AI reads cells

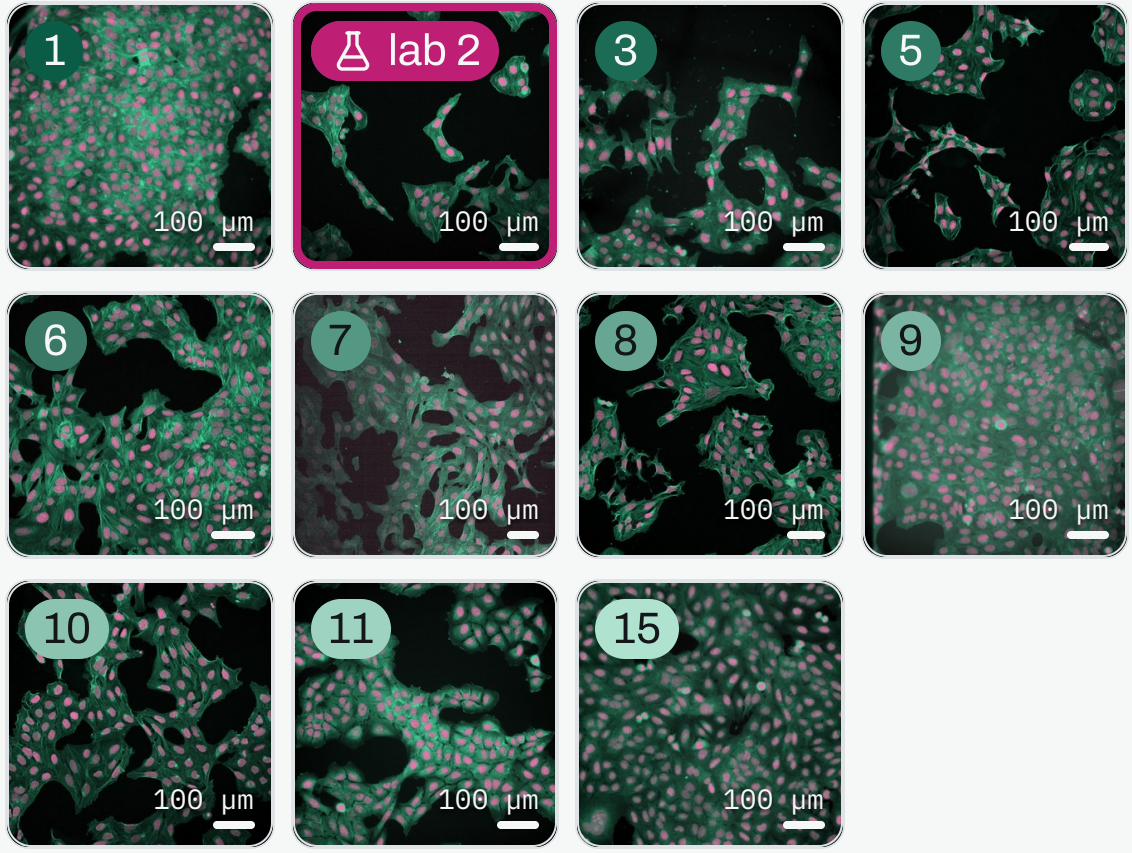
One lab, or eleven?
Guess out loud.

tap a tile to check

✓ same cells: **U2OS** ✓ same protocol: **Cell Painting** ✓ **DMSO** vehicle control: solvent only, no drug

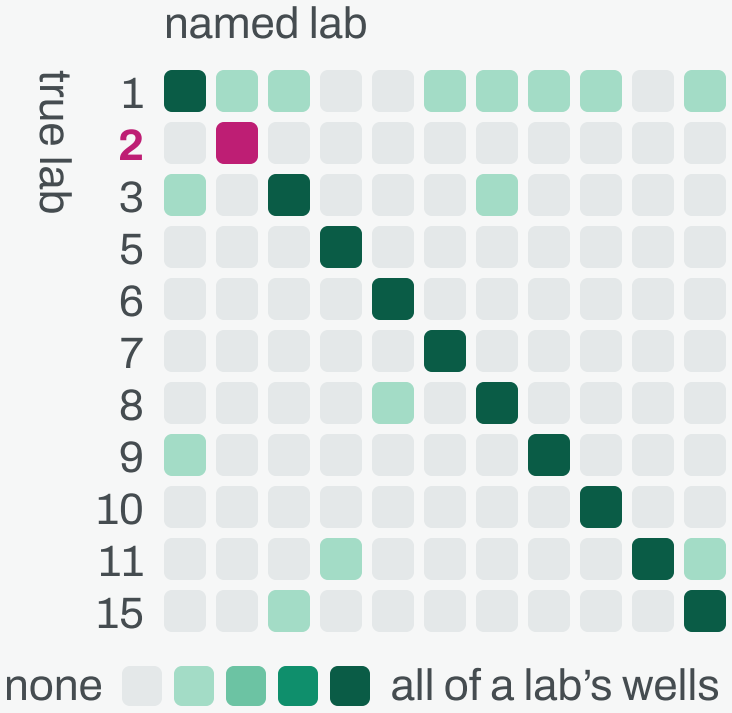


Could anyone tell?
A classifier can.



99.9%

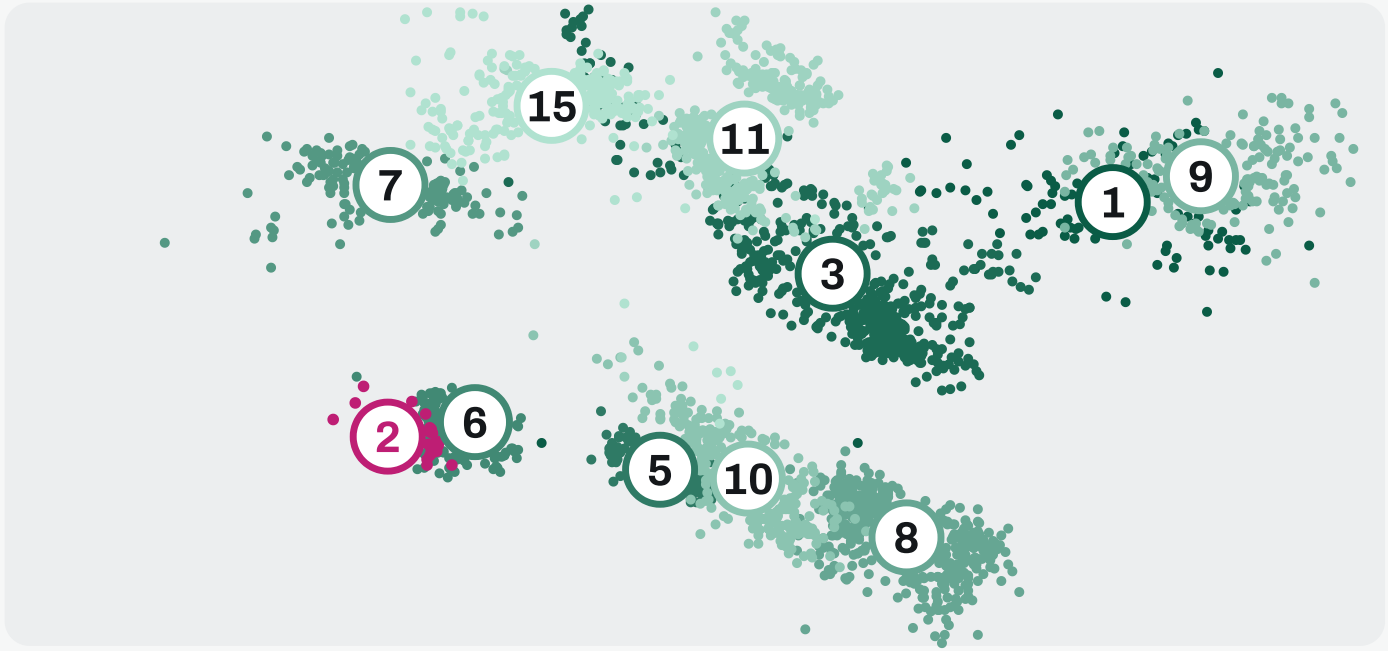
right lab, 93,228 wells
chance: 1 in 11



Biosecurity: labels can lie, cells cannot.

WHITE PAPER ↗

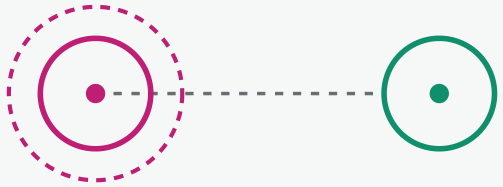
Eleven labs.
Eleven “normals”.



one well (15,352) a lab’s “normal” closer = more alike

6×

between two labs



lab 2 another lab

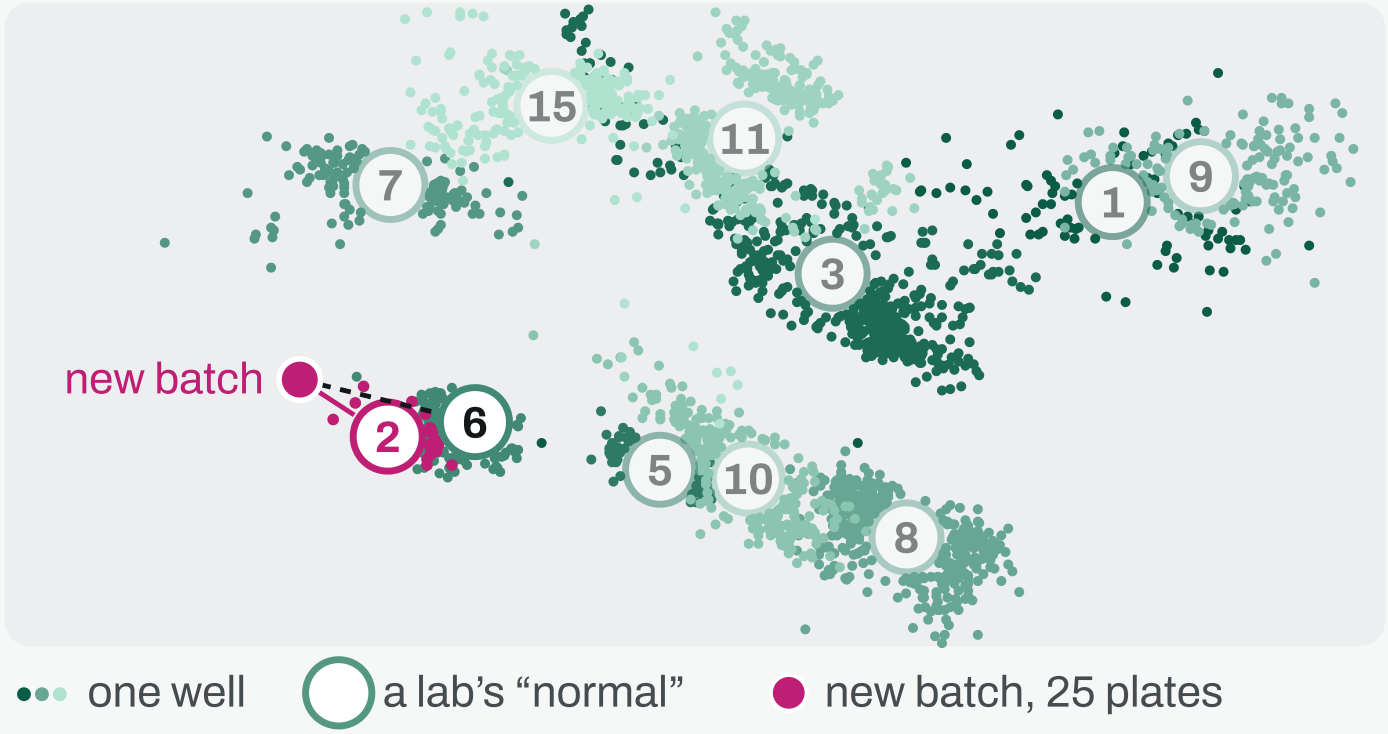
- one well: ~220 cells

1× wells on one plate

1.6× a lab’s batches

WHITE PAPER ↗

Send the controls.
Know it’s “normal” before the screen.



Member check

a new batch, in all features

own lab	**2.8×**
next lab	4.4×

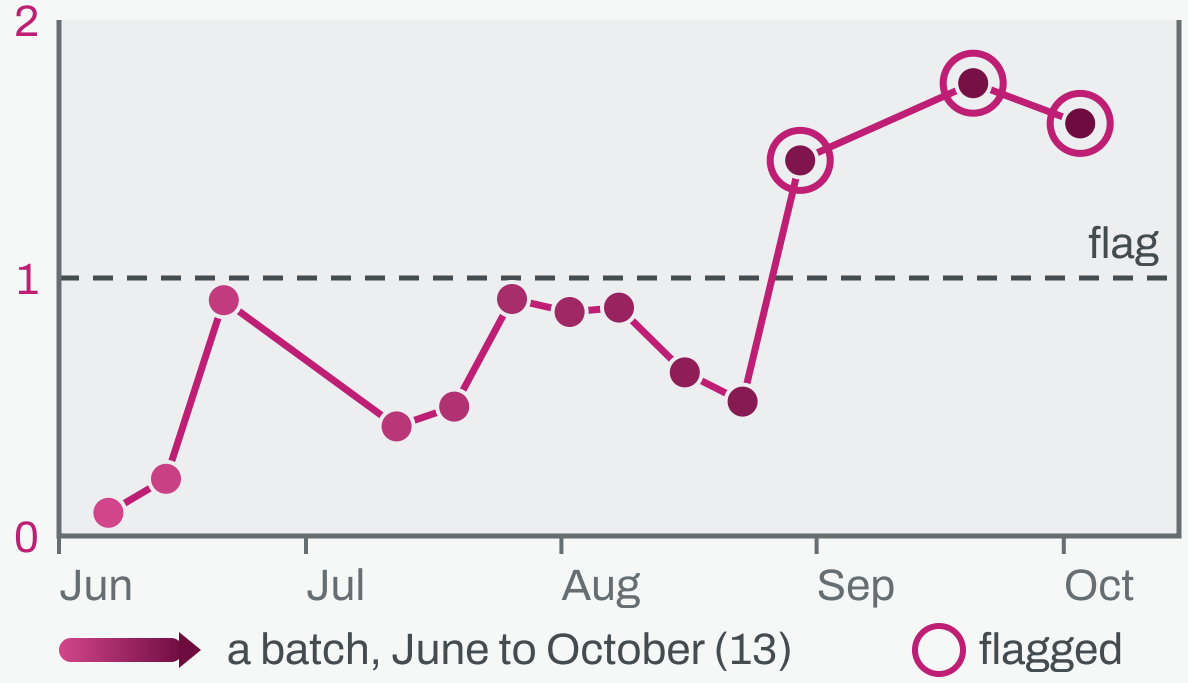
- nearer its own lab ✓
- drugs match ✓
- grew as usual ✓

WHITE PAPER ↗

Controls catch drift before any drug is read.

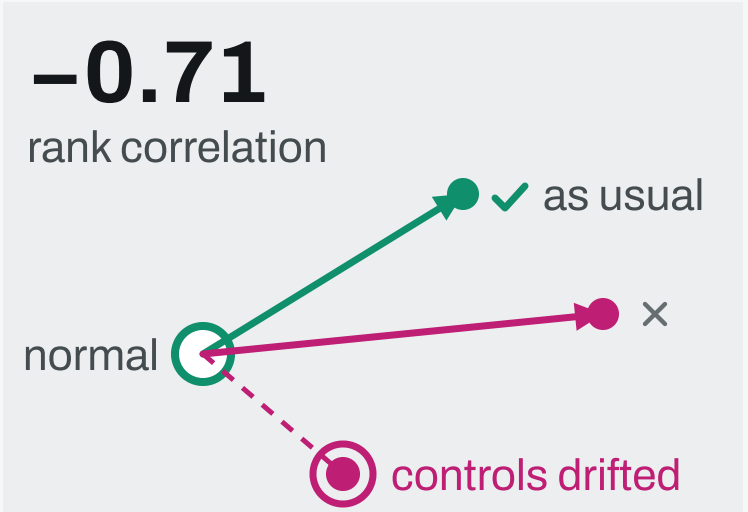
lab 2's controls, drift

1 = as far as another lab



a drug's answer, read from normal

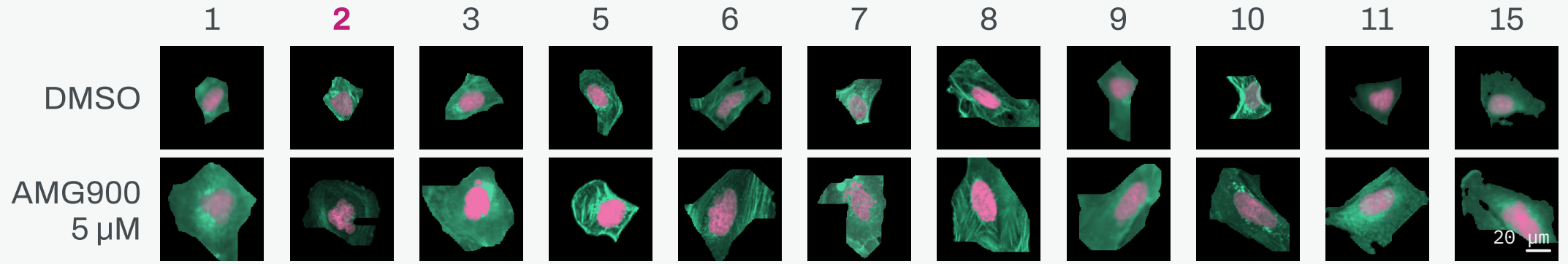
the batch drifts, the answer turns

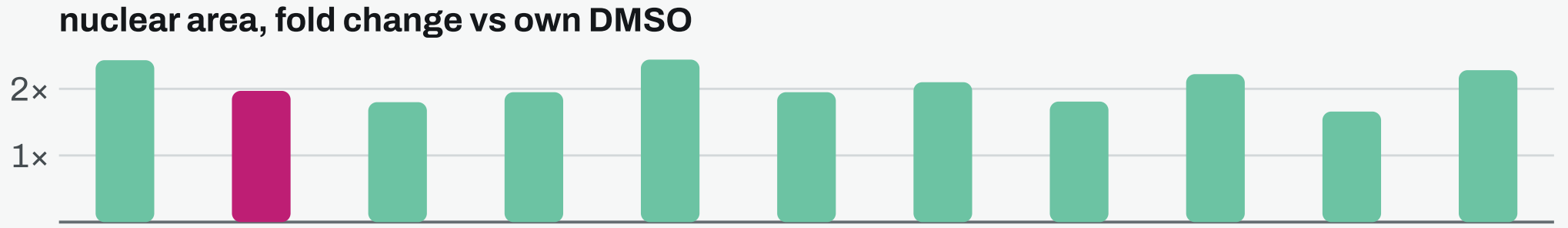


× known drug fails

PRODUCT VISION ↗

The product is the data.
One drug: nuclei ×1.7 to 2.4 in all 11 labs.





PRODUCT VISION ↗

11 labs' public images, plus yours.
Only control wells are shared.

SOURCES

public images
JUMP, 11 labs, open

your raw images
backed up free, off-site
the copy insurers ask for

→

one repository

shared
control wells, on one map

private
everything else, yours alone

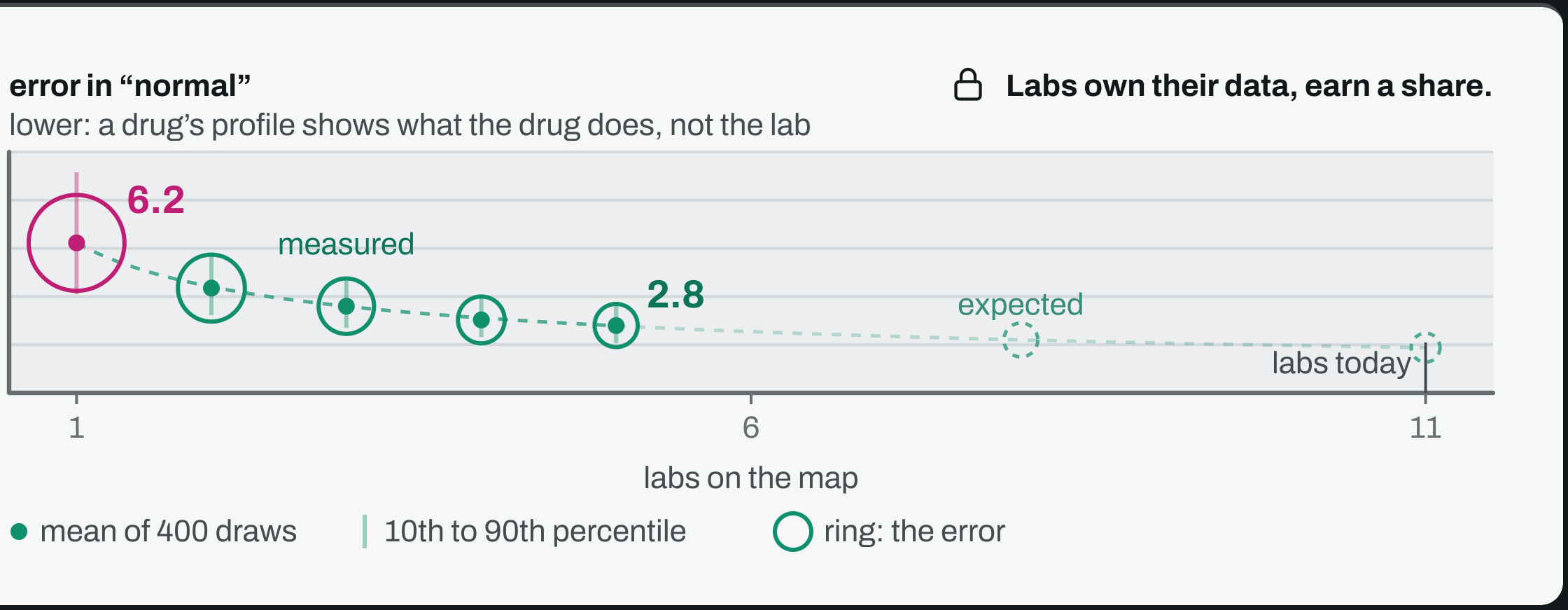
→

MEMBERS

screening teams
a check on every batch

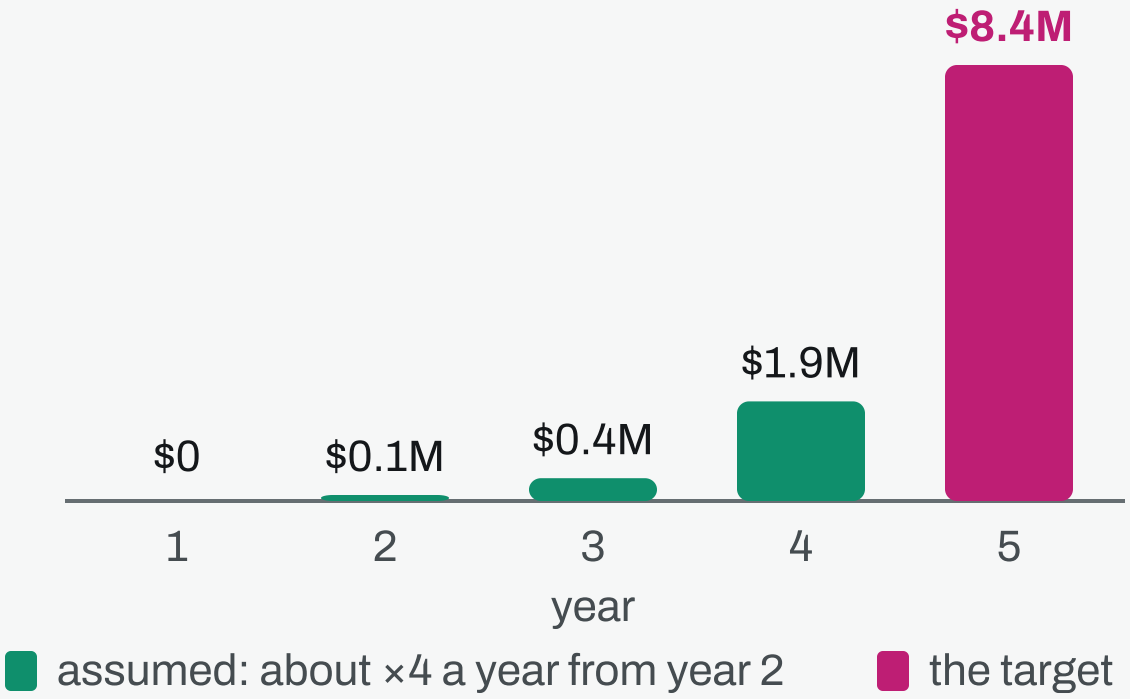
AI teams
aligned images, by API

Every lab that joins sharpens “normal”: error 6.2 with one lab, 2.8 with five.



Every likely buyer: $84M/year.
Five-year target: a tenth, $8.4M/year.

five-year plan, revenue a year



who pays, every likely buyer



a tenth: $8.4M/year

data contributors	free access	$0
screening labs	$10k/line/quarter	$64M
buyers of cell data	$25k/dataset	$15M
big screening panels	$150k/year	$3.75M
benchmark partners	$100k/year	$1M

LANDSCAPE ↗

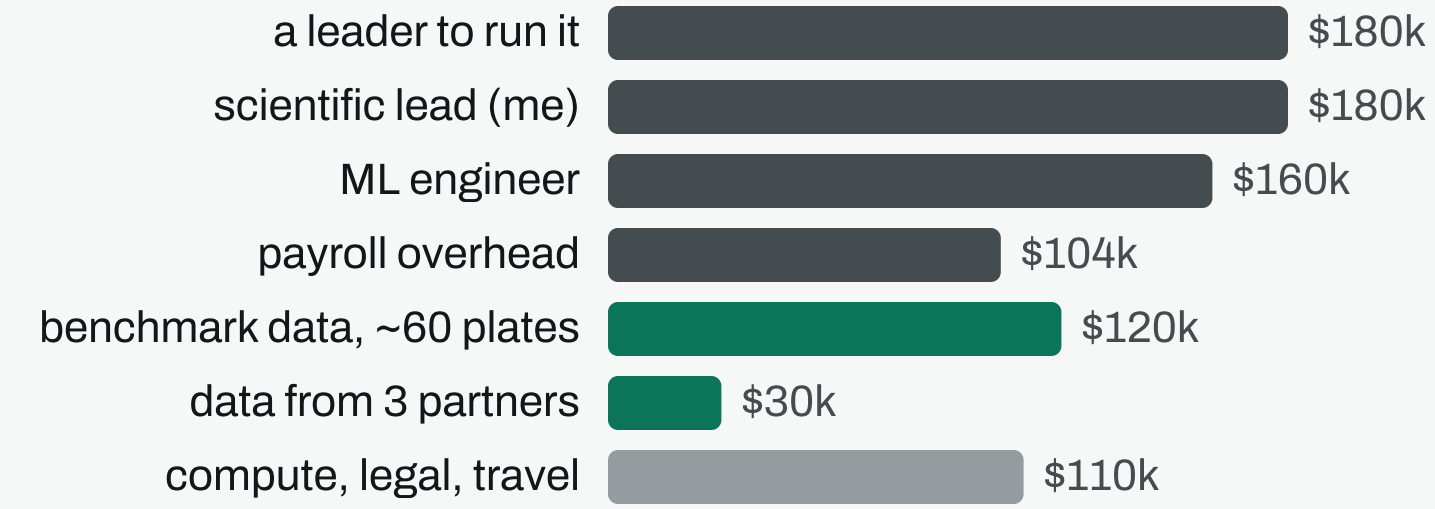
Each tool sees one lab.
Datum Line sees every lab.

	measure cells	normalise plates	reference data	**every lab, over time**
CellProfiler	✓			
Harmony	✓			
pycytominer		✓		
Genedata Screener		✓		
JUMP, Recursion			✓	
Datum Line				✓

Datum Line hosts every lab's raw control images, and runs on them.

ROADMAP ↗

Raising $1.8M pre-seed for 18 months: first paying members, then the seed.



$884k plan + $884k contingency = $1.77M (A7)

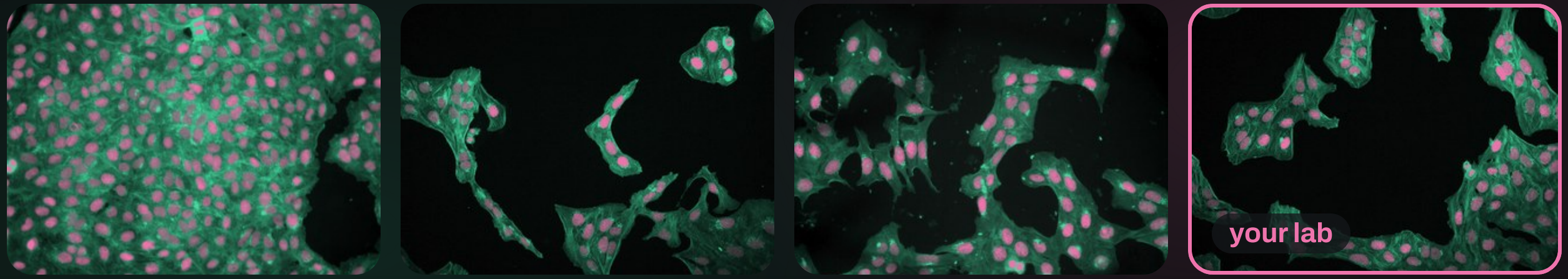
TODAY

- **11 labs, 129 batches mapped** from public data; no revenue yet

BY MONTH 18

1. **first data outside JUMP**
2. **first paying member**
3. **benchmark playground**
4. **model for any cell**

→ seed round



There’s a tile for your lab.
Let’s map “normal” together.

$1.8M
pre-seed for 18 months

Have control images? Talk to me.

KAT-TITTER.GITHUB.IO/DATUM-LINE

The rest, for questions.

Every number: the code here or a cited source.
All 25 boards: archive.html
Documents: white paper, product vision, roadmap, landscape

A1 **Why not batch-correct it away?**
erases most of the lab fingerprint

A2 **Does it change the result?**
same drug, different answer

A3 **Does it work without stains?**
bet on brightfield

A4 **What is private?**
control wells shared, nothing else is

A5 **What is the machine learning?**
image in, position and correction out

A6 **What about biosecurity?**
cells show their lab, labels can lie

A7 **How much, and what for?**
planned $0.9M × 2 = $1.8M asked

A8 **How big?**
count × price, three streams

A9 **Why would a lab share?**
owned by the lab, verified by anyone

The usual correction erases most of the lab's fingerprint.

JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

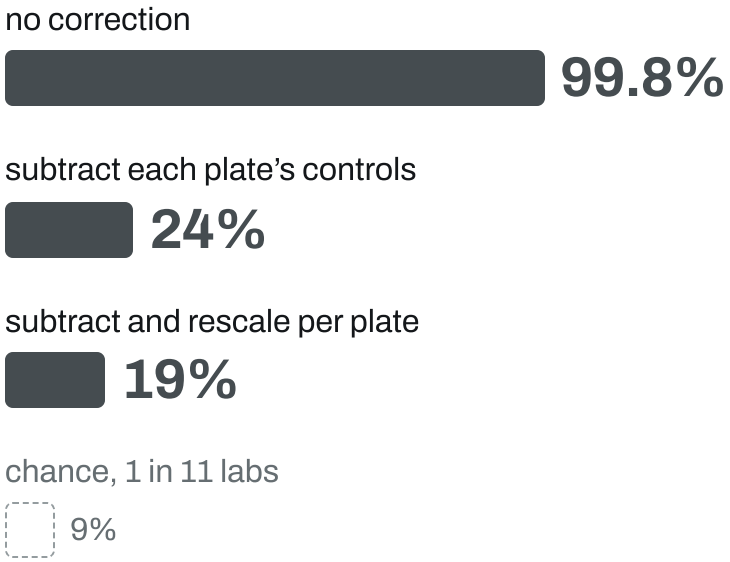
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



29,943 DMSO wells, 16 per plate
tested on plates it never saw

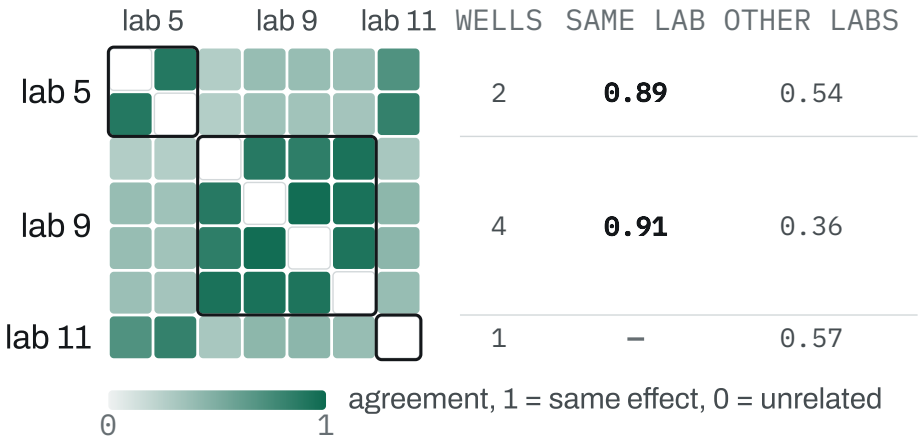
Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

The same drug looks less alike in another lab.

Darker = more alike.

WHAT A DRUG SCREEN REPORTS — drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER



	WELLS	SAME LAB	OTHER LABS
lab 5	2	**0.89**	0.54
lab 9	4	**0.91**	0.36
lab 11	1	–	0.57

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

	SAME LAB	OTHER LABS
RSL3, 10 µM	**0.90**	0.49
8 known drugs, 5 µM, median	**0.74**	0.26
DMSO vs DMSO	**0.00**	0.00

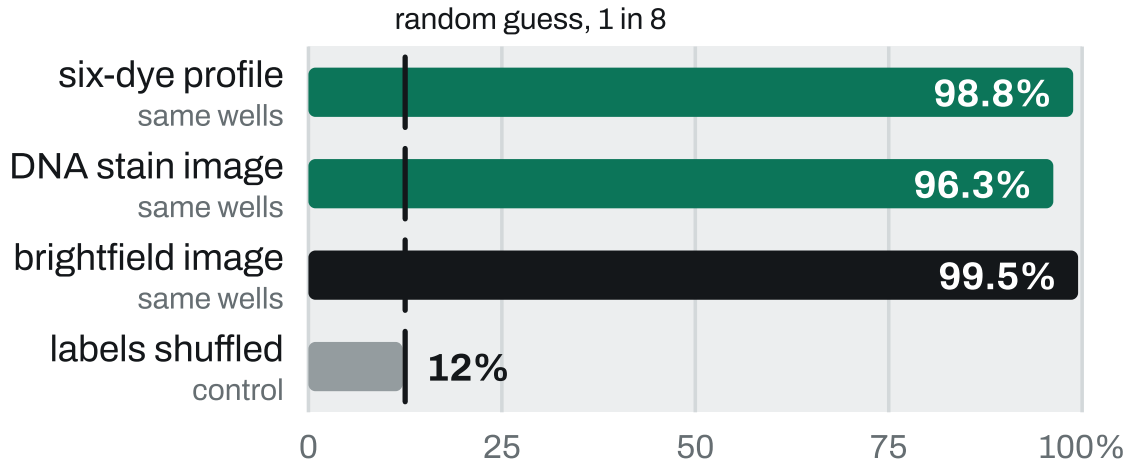
DOES IT CHANGE THE NUMBER?

Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

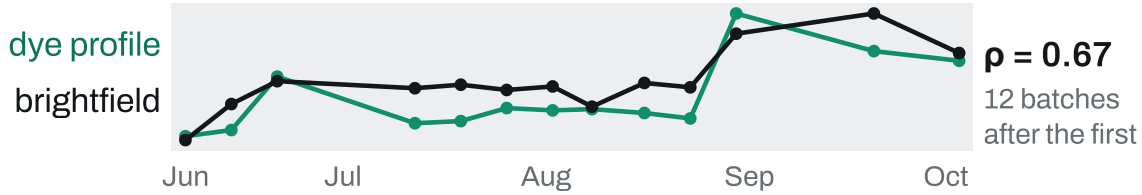
One brightfield image already names the lab.

Evidence: six dyes.
Product: no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?



DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

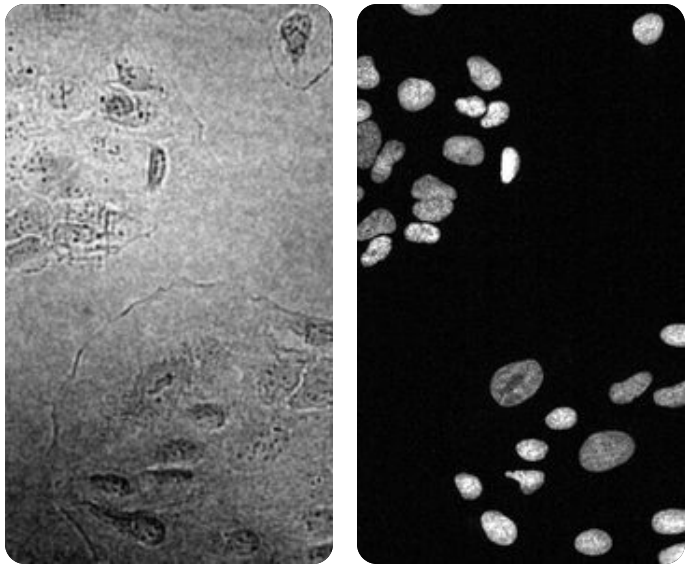


Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A1



BRIGHTFIELD DNA STAIN

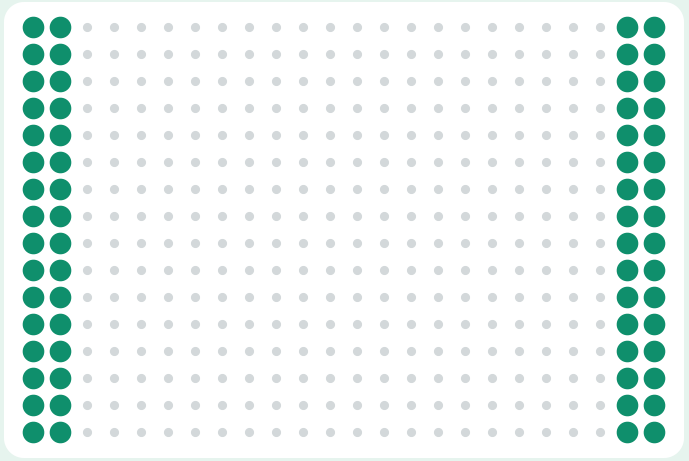
same spot in one well, lab 2

Only control wells are shared.

Proposed: a free backup of the rest, encrypted with **your key**.

WHAT IS SHARED

control wells only: DMSO and known drugs



64 of 384 wells

images, or numbers from them
proposed: images deleted once measured

a plate map and the batch date
which wells, and when they were imaged

WHAT DATUM LINE DOES WITH IT

the map holds DMSO wells only

82,319

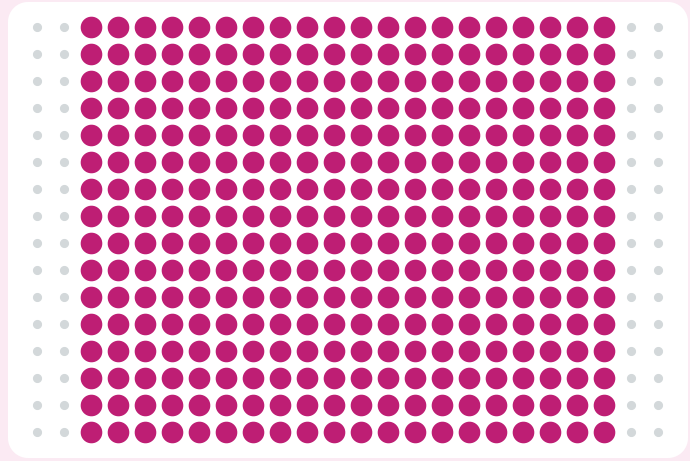
DMSO wells of 10 other labs
0 treated

builds the map
scale set by the other labs, never its own

answers
“normal” or not, how far it moved, nearest lab

WHAT IT NEVER READS

your treated wells



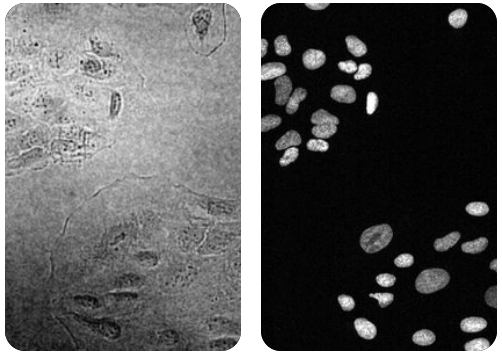
320 of 384 wells

your results
corrected in your building, by you

your name, unless you choose
proposed: a number until you opt in

An image goes in; a position and a correction come out.

1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

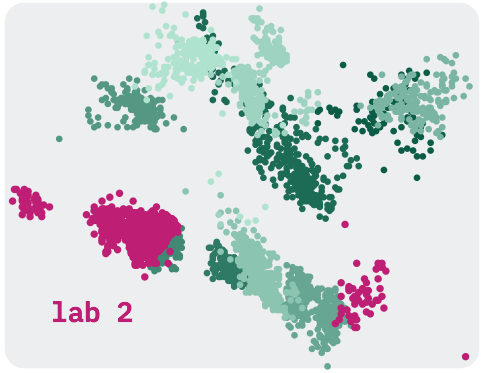
2, NUMBERS

3,773

numbers per well

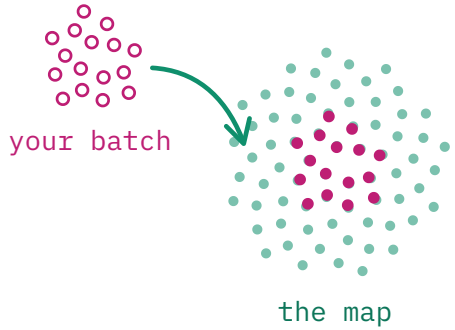
standard software today: size, shape, brightness, texture

3, THE MAP



each lab placed on the other labs' scale, never its own

4, THE CORRECTION



one per batch, fitted on your DMSO wells, sketch, archive A14

AI IMAGE MODEL NEXT, A3

brightfield, 8 labs 99.5%

DNA stain, 8 labs 96.3%

right lab, from one picture

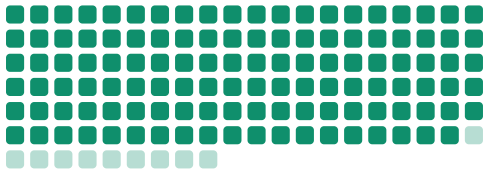
Every lab leaves a fingerprint, so a wrong label stands out.

Two work today, three are proposed.

PROVENANCE today

99.9%

of 93,228 DMSO wells get the right lab, on unseen plates: cells and microscope together.



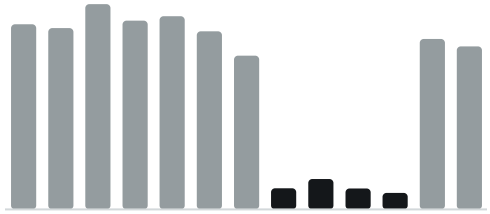
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



lab 3, drug effect per batch
dark: failed, archive A8

UNKNOWN ORIGIN proposed

9×

Farthest of 129 real batches from its nearest other lab.



distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

30 Aug

2021: lab 2's DMSO wells moved; its cell count did not.



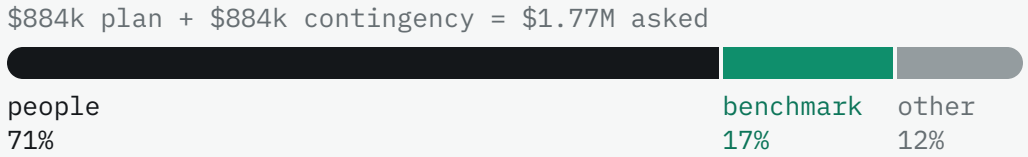
lab 2, since batch 1
filled: flagged, archive A13

DECLARED VS OBSERVED

proposed, opt-in

A company that wants the good-actor mark sends treated wells and says what is in them. The map flags cells that look like a known toxic class [7]. Only what a lab sends is seen.

About $1.8M for eighteen months, mostly people.



leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates, planted faults, archive A16	**$120k**
partner-lab bench time	3 labs, controls they already image, reagent top-ups, shipping	**$30k**
compute and storage	images, models, hosting, API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners, 3 labs	**$20k**
planned	sum	**$884k**
+ contingency	most things take 2–3× the time and money planned, we plan for 2×	**$884k**
asked, 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

1 data from three partner labs, the first outside JUMP, paid in this round **2** first paying member
3 benchmark playground **4** model for any cell → seed

About $84M a year, if every likely buyer joined.

assumption	low	base	high
organisations testing drugs on a few cell lines	200	400	800
cell lines each one uses per quarter	2	4	8
price per cell line per quarter, $k	5	10	15
organisations testing drugs on hundreds to thousands of cell lines	10	25	50
site licence per year, $k	50	150	300
per-line checks and site licences, per year	**$8.5M**	**$67.8M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each places on the map per year	2	4	8
price per dataset membership, $k	10	25	50
dataset memberships on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
total per year, if every buyer joined	**$9.7M**	**$83.8M**	**$523M**

BASE CASE, PER YEAR



WHY BUYERS PAY

A line checked late, in a screen I ran:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
A $25k dataset membership: about 1% of that.

The lab keeps its data and a share; anyone can check it.

No crypto token.

CONTROL WELLS ONLY — today

no test compound, no target, no result
is ever shared or read

LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

KNOWN MEMBERS — proposed

verified at sign-up, odd access flagged
access log open to members

OPEN CODE AND DATA — today

code: MIT, on GitHub
data: JUMP, CC0; every number from a script

